

Weekly Netlist Changes Report

Voltage Sources

This week, I introduced null (zero-volt) voltage sources to access circuit branch information not natively accessible in `ngspice`. This technique allows for current measurements at previously unobservable nodes. The circuit now utilizes five voltage sources in total: two pulse sources, one primary supply (`vdd`), and two 0V sources dedicated to current measurement.

Obtaining C_{in}

Gate capacitance was calculated using three different methodologies:

RC Circuit

The first method models the transistors as an RC circuit by connecting individual 100 k Ω resistors to both the NMOS and PMOS transistors. Using independent resistors prevents the detection of undesired voltages and improves measurement accuracy. The resulting capacitance difference between the NMOS and PMOS devices now appears much more realistic than previous measurements, which were improbably similar.

As expected, the PMOS exhibits a higher gate capacitance due to its larger size. While using resistors as large as 100 k Ω yields realistic capacitance values consistent with the other methods, the exact theoretical basis for this effect requires further investigation.

Integral

The second method determines gate capacitance using the standard capacitor equation, $C = Q/\Delta V$. For our application, this is adapted by integrating the current over time: $C = \frac{\int I dt}{\Delta V}$

This calculation was performed individually for both the PMOS and NMOS transistors. This approach also motivated changes to the voltage sources. Previously, energy was obtained by integrating the total current through a shared voltage source feeding both the PMOS and NMOS. By introducing separate null voltage sources, the energy for each transistor can now be integrated individually. This refinement produced a more realistic divergence between the NMOS and PMOS capacitance values.

Equation

The third and final method calculates gate capacitance using the canonical oxide capacitance equation: $C_g = C_{ox} \times W \times L$. Unlike the previous empirical methods, this approach relies directly on parameters provided by the transistor model file.

Method	PMOS (fF)	NMOS (fF)
RC Delay	0.8138	0.3975
Integral	0.6664	0.3255
Equation	0.5204	0.2729

Average C_{in}

By convention, our upcoming simulations will use a load capacitance (C_{out}) equal to four times C_{in} . To determine a unified C_{in} value, the NMOS and PMOS capacitances from each method were summed to find the total gate capacitance for each technique. Averaging these three totals yields a final mean C_{in} of 0.9809 fF.

Delays

The methodology for calculating delays is well-established and relies on straightforward concepts, requiring no further methodological adjustments. The following delay metrics (Table 2) were simulated using a total C_{load} of $4 \times C_{\text{in}}$ (3.9236 fF).

td (ps)	tdlh (ps)	tdhl (ps)	tf (ps)	tr (ps)
21.78	23.37	20.19	32.10	44.11

Power Consumption

The most significant progress this week involved a deeper understanding of transistor power consumption dynamics. First, the instantaneous power consumed by the vdd source was calculated using $P = I \times V$. Integrating this power over the entire transient simulation yields the total energy consumed. Dividing this total energy by the simulation time provides the average total power.

For dynamic power, I isolated both the short-circuit and the C_{out} load power. By observing the current through vdd at specific transition points, the short-circuit current is isolated during the 1-to-0 output transition, whereas the 0-to-1 transition displays the sum of the short-circuit power and the load power.

By pinpointing the exact timing of these current spikes, I separated the short-circuit power from the load power. I calculated the power consumed during the first spike (which consists solely of short-circuit power) and multiplied it by two to estimate the total short-circuit consumption per cycle. Because the subsequent spike represents the sum of both power types, I subtracted the isolated short-circuit power from this total to determine the load power.

Notably, this empirical approach circumvents the need for the theoretical formulas provided in the initial research group reference article. Despite minor discrepancies, the extracted values are highly satisfactory:

Total Power (μW)	Short-Circuit Power (μW)	Load Power (μW)
7.96	4.95	3.36